Kush Deoghare

Seeking a 4-6 Month Master's Internship in Embedded Systems Engineering

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Highly motivated Electronics and Embedded Systems Engineer with strong technical expertise in microcontroller programming, FPGA/VHDL design, IoT integration, and robotics. Proven ability to deliver innovative embedded solutions through hands-on project implementation and applied research. Combines technical depth with demonstrated leadership, cross-cultural collaboration, and customer-facing experience gained through diverse professional roles. Seeking to contribute to cutting-edge embedded systems development in an industry internship.

EDUCATION

ESIGELEC, Rouen. France (in partnership with INSA Rouen), 09/2024 – Present
MSc in Electronics and Embedded Systems Engineering France

- Currently involved in an R&D project on an autonomous drone using ROS2, ORB-SLAM–based concepts, and computer vision for real-time perception and navigation. Implemented gesture-based drone control using MediaPipe hand tracking with temporal confidence logic. Integrated face recognition using Haar Cascades combined with TensorFlow / Keras for identity verification. Designed a custom monocular visual SLAM (V-SLAM) pipeline using a single RGB camera to estimate free space and infer candidate exits in unknown indoor environments.
- Developed expertise in ARM Cortex-M and MSP-430 microcontroller programming using Embedded C for real-time applications
- Designed FPGA and VLSI circuits using VHDL and Verilog for hardware-software integration
- Implemented IoT systems with wireless protocols (Bluetooth, Zigbee, Wi-Fi) for connected devices
- Developed applications using Real-Time Operating Systems (RTOS) for multitasking embedded systems
- Implemented Python-based image processing and computer vision algorithms

Heriot-Watt University, Edinburgh, UK, 2019 – 2023
Bachelor of Engineering in Computing & Electronics Engineering Edinburgh, UK

- Comprehensive foundation in digital/analog electronics, embedded systems, computer architecture, control systems, AI, and computer vision
- Developed programming expertise in C, C++, Python, and MATLAB for embedded systems and robotics
- Completed capstone project applying multidisciplinary engineering skills to integrated system design

PROFESSIONAL EXPERIENCE

Mindrift, *Freelance AI Tutor/ Trainee* 10/2025 – Present
Remote

- Contributed domain expertise in electrical engineering to train generative AI models, crafting prompts, generating responses, and evaluating outputs for accuracy in signal processing, power electronics, and microelectronics applications.
- Reviewed and refined AI-generated content on technical topics like DSP algorithms, power converter design, and microelectronic circuits, developing conversational examples and scoring rubrics to enhance model reliability.
- Annotated training data for specialized EE projects using knowledge of embedded systems, analog/digital signal processing, and power systems, ensuring AI outputs support real-world engineering applications.
- Delivered flexible, high-quality human input on-demand, adapting to project requirements while upholding ethical standards and technical precision.

Gamers Incorporated, *Technical Support Specialist* 2018 – 2019
Remote

- Provided remote technical troubleshooting and problem resolution for international clients
- Developed strong analytical and customer communication skills in technical contexts
- Demonstrated ability to diagnose complex technical issues and deliver effective solutions

TECHNICAL COMPETENCIES

Embedded Systems & Microcontrollers — ARM Cortex-M, STM32CubeIDE, MSP430, Arduino, Raspberry Pi, BeagleBone | Embedded C, Real-Time Systems, Hardware-Software Integration

FPGA & Digital Design — VHDL, Verilog, Intel FPGA Development | Digital Logic Design, Circuit Simulation, System-on-Chip Architecture

IoT & Communication Protocols — I2C, SPI, UART, Bluetooth, Zigbee, Wi-Fi, M2M Communication, Wireless Sensor Networks, Connected Devices

Software Development — C, C++, Python, Java, MATLAB, Object-Oriented Programming, Algorithms, Version Control (Git), Kotlin, Android Studio, Android SDK, Android app development

Electronics & Control Systems — Digital/Analog Circuit Design. PCB Design & Schematic Capture, Power Electronics, Sensors & Actuators, PID Control, Autonomous Navigation. Robotics Control Systems

Development Tools & Platforms — Keil uVision. Code Composer Studio, MATLAB/Simulink, Linux (Embedded). RTOS (uC/OS-II) Oscilloscope, Logic Analyzer, Debugging Tools, Xilinx, Microsemi

Specialized Skills — Computer Vision: OpenCV, ORB feature extraction, monocular V-SLAM, MediaPipe, Machine Learning: TensorFlow, Keras, Haar Cascade–based face detection, Network Design (Cisco), Web Development (HTML, CSS, PHP, JavaScript), ROS2

TECHNICAL HIGHLIGHTS

FPGA & Digital Systems
- Designed UART transmission receiver with interested laser communication for real-time data transfer
- Implemented digital logic designs on Intel FPGA Starter Development Kit with circuit simulation

Embedded Systems Development
- Built line-following robot with MSP430, integrating sensors, motor control, and PID-based navigation
- Developed multi-tasking RTOS application on MSP430 with μC/OS-II and Sharp96x96 display
- Programmed MSP430 Booster Pack MK-II for hardware-software integration tasks
- Created embedded Linux applications on BeagleBone platform with full system integration
- Autonomous drone navigation using ROS2, ORB-SLAM concepts, monocular V-SLAM, MediaPipe gesture control, and vision-based face recognition

IoT & Communication Systems
- Developed multi-device communication system combining I2C and SPI protocols using Arduino and MSP430
- Designed M2M communication protocols for connected device networks
- Built FM radio from scratch, including circuit design, PCB layout, soldering, and hardware integration

Sensor Systems & Signal Processing
- Implemented NADPCMC algorithm in Python for advanced sensor data processing and analysis
- Developed image processing applications in Python using OpenCV and Pillow for sensor and vision data.
- Implemented DSP algorithms in C/C++ and MATLAB, gaining practical experience in filtering, sampling and signal analysis for embedded systems.
- Developed and tested embedded applications on STM32 Discovery boards using STM32CubeIDE and HAL libraries.

Software & Application Development
- Built Library Management System with full CRUD functionality and user authentication (HTML, CSS, JavaScript, PHP)
- Designed comprehensive network system using Cisco Packet Tracer with routing and protocol configuration

Mobile and Android Development
- Developed a native Android app that retrieves song lyrics from online sources • Integrated Apple iTunes Search API to stream legal 30-second audio previews • Implemented REST API handling, JSON parsing, and asynchronous networking

KEY COMPETENCIES

Technical — Embedded Systems Design, FPGA Programming, IoT Integration, Real-Time Systems, Digital Circuit Design, Microcontroller Programming, Hardware Debugging, System Architecture, Bash, Make, CMake • **Soft Skills** — Project Management, Team Leadership, Cross-Cultural Collaboration, Technical Documentation, Problem-Solving, Customer Communication, Quality Assurance, Attention to Detail, Adaptability

LANGUAGES

English — Fluent (Native proficiency) • **French** — Intermediate (B1-B2) • **Hindi** — Native • **Marathi** — Native